

Design and implementation of a Portable Low-Cost Internet-Enabled I - V Curve Tracer Utilizing Cuk Converter

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Keywords: CUK CONVERTER, I-V TRACKER, PORTABLE, LOW COST

Abstract

This work presents the development of an internet-enabled current-voltage (I - V) curve tracer prototype circuit. With an emphasis on cost-effectiveness and data accessibility, the prototype was constructed utilizing Raspberry Pi 4 as the data acquisition board equipped with a TFT display and webserver connectivity. The prototype was designed to handle up to 60V open-circuit voltage, 10A short-circuit current and a maximum power point of 250W. The measured data is processed programmatically with a filtering algorithm to smoothen the curve by removing outlier values measured due to noise. Further refinement was achieved by averaging multiple data points recorded at each step of a single duty cycle change. The prototype was tested on three commercially available PV modules through Chroma 62100H-600HS Solar Array Simulator. The proposed device achieved an overall average RMSE and RRMSE of 0.2848A and 6.4985% respectively. By combining linear duty cycle with linear gain changes across different regions of the I - V curve, a more even distribution of data points was achieved, resulting in a total scan time of 6.8 seconds. While sufficient for less accuracy-intensive tasks, the current prototype is not ideal for research-grade applications. It is expected that with the use of a dedicated printed circuit board (PCB), for reduced noise and enhanced signal integrity, higher accuracy could be achieved. Nonetheless, this project successfully demonstrated the feasibility of incorporating internet connectivity for remote monitoring of PV module status.

1. Introduction

Unlike traditional energy sources, green energy harvesting methods became increasingly popular in recent years. More and more countries and unions have begun to transform the traditional method into sustainable alternatives. One of the most widely adopted methods is using photovoltaic (PV) modules to convert sunlight into electrical energy. According to [1], solar energy capacity around global increases from 140 GW (2013) to 1053 GW (2022). Moreover, the capacity of annually installed PV systems would reach 3800 GW by 2030 [18].

As the number of PV modules continues to increase annually, it is essential to monitor and maintain the state of PV modules to identify issues and maximise efficiency. A current-voltage (I - V) tracker is the most effective tool for monitoring the I - V and power-voltage (PV) characteristics of solar PV modules. These characteristics are valuable not only for the optimization of power converter and PV system designs, but they are also widely used for diagnosing the health and performance status of PV modules [19,9].

There are several methods for tracking I - V characteristics, starting with the simplest and most cost-effective: the impedance-based method. This approach uses a variable resistor to alter the load and measure the resulting current and voltage. However, it has limited accuracy and requires manual adjustment, making it unsuitable for real-time monitoring. Additionally, this method is less efficient due to power dissipation as heat. [2,3]

In addition, MOSFET- and capacitor-based I - V tracers are also cost-effective options, though their performance depends heavily on the quality of the components used. [6,16] Capacitor-based tracers, however, present challenges in data processing due to the dependence on sweep speed, number of samples, and switching dynamics. Additionally, this method requires precise timing and differentiation to derive current, which leads to longer computation times. As a result, such tracers are unsuitable for automation. [6] On the other hand, MOSFET-based tracers are simpler to automate due to their moderate complexity. However, they offer only moderate efficiency and result in low cycle count due to their relative high heat dissipation during operation [16].

On the other hand, the DC/DC converter-based approach stands out as one of the most efficient techniques for I - V curve tracing. By precisely adjusting the gate voltage of MOSFET, DC/DC converters can efficiently sweep through a wide range of voltages and currents, providing high accuracy across the entire curve [3]. With the integration of a Raspberry Pi, this method supports real-time data collection and can handle dynamic environmental changes. However, the method tends to be more costly due to the need of sophisticated components, such as high current rating inductors, power MOSFET and control systems.

From the perspective of data management, conventional I - V curve tracers are offline devices which stores data locally, with the option to display the measurements on an in-built display. This implies that data analysis must be performed offline after measurements. Due to the harsh environmental conditions the PV systems are typically deployed under, the ability to communicate the measurement data online would be

advantageous as it allows real-time monitoring and data analysis activities to be performed remotely.

In view of the discussion above, this work presents the design and implementation of a portable and low-cost internet enabled I - V curve tracer based on the Ćuk converter design. The prototype is designed to operate within the open-circuit voltage (V_{oc}) range and short-circuit current (I_{sc}) range of 0-60 V and 0-10 A, respectively. The design emphasizes on performance, cost-effectiveness, and usability. The converter components are chosen for complete sweep of I - V curve and P-V curves. The measurements are stored on a cloud platform, which allows engineers and researchers to view real-time I - V data without the need for physical presence at the testing site. Moreover, with access to the historical data stored, the performance trends of the solar panels are readily analysed and modelled for improved modelling and prediction studies.

This paper is organized as follows: First, the background theory on the design of the I - V curve tracer is presented. Next, the implementation of the proposed design is discussed. It was tested up to 250W at maximum power point (MPP) using Chroma 62100H-600S Solar Array Simulator. The results section deliberates the performance of the prototype for various PV modules and environmental conditions.

2. DC-DC Converters

2.1 DC-DC Converters

The dc-dc converter method allows precise and smooth variation of emulated resistance seen by the solar cell without significant power loss. The three basic DC-DC converter topologies are the buck, boost, and buck-boost converters. The varying of the output voltage can be related by the DC gain. The desire output voltage can be achieved by adjusting the duty cycle, d within the range of 0 to 1.

$$V_o = \text{gain} \times V_{in} \quad (1)$$

With the assumption that all power supplied by the PV generator is consumed by the load [4], it follows that:

$$V_{in} I_{in} = \frac{V_o^2}{R_o} \quad (2)$$

By substituting (1) and (2),

$$R_{eq} = \frac{V_{in}}{I_{in}} = \frac{R_o}{\text{gain}^2} \quad (3)$$

Equation (3) explains that the equivalent resistance of the dc-dc converter seen by the PV generator is the ratio of output resistor R_o to gain squared.

Table 1 Gain and Equivalent Resistance of DC-DC Converters

DC-DC Converter	DC Gain	Equivalent Resistance, R_{eq}
Buck	D	$\frac{R_o}{D^2}$

Boost	$\frac{1}{1-D}$	$(1-D)^2 R_o$
Buck-boost (Zeta, Ćuk, SEPIC)	$\frac{D}{1-D}$	$\frac{(1-D)^2 R_o}{D^2}$

To achieve a complete sweep of I - V curve from open-circuit voltage to short-circuit current, the converter must produce an infinite equivalent resistance at 0% duty cycle and zero equivalent resistance at 100% duty cycle, representing open-circuit and short circuit condition respectively. From Table 1, only the buck-boost derived converters can satisfy the conditions.

The buck-boost-derived topologies such as the Zeta, Ćuk and SEPIC converters, are commonly used for the characterization of the photovoltaic panels. All the topologies offer similar function to direct current transformation, by stepping up or stepping down the input voltage. The Zeta converter features discontinuous input current, leading to high input ripple and noise issues. In contrast, Ćuk and SEPIC converters feature non-pulsating input currents, allowing for a less noisy I - V measurement. While Ćuk and SEPIC topologies shares similar characteristics and circuit layout, the Ćuk converter has an inverting output voltage relative to the input.

Given these characteristics, Ćuk converter was chosen as the dc-dc converter used for the I - V curve tracer, offering stable input current and the ability to go from open-circuit to short-circuit condition. The inverting output voltage is not a concern in the case of this I - V curve tracer.

2.2 Ćuk Converter

Like other types of dc-dc converter, Ćuk converter operates on the principles of energy transfer between inductors and capacitors, producing a step-up or step-down voltage at the output. The converter is made up of two inductors (input side L_1 , output side L_2), a coupling capacitor C_1 , a switching device (MOSFET or transistor), a diode for current rectification, an input and output capacitors (C_{in} and C_{out}).

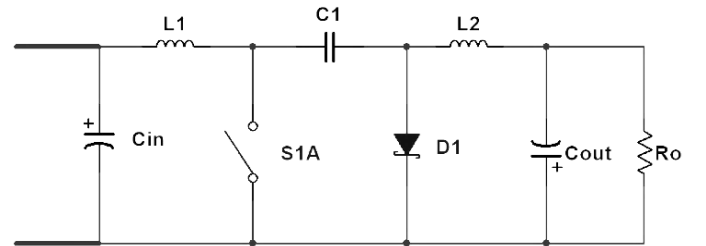


Figure 1: Ćuk Converter Circuit Diagram

The Ćuk converter operates in two main states: switch-on-state and switch-off-state. The transition between these two states is controlled by the switching of the MOSFET. The converter's operation can be understood in four steps:

1. When the MOSFET is first turned on, current flows through input inductor (L_1), completing the circuit. Energy is stored in the form of electric field in L_1 .

2. When the MOSFET turns off, L_1 releases its stored energy to the coupling capacitor (C_1), charging it and the circuit is completed through the diode.
3. The MOSFET turns on again, allowing L_1 to charge once again after discharging in the previous cycle. The charged C_1 now act as an energy source, supply current through the conducting MOSFET to the output capacitor (C_{out}) and the load resistor (R_o). The current returns to the other side of C_1 via the output inductor (L_2), while charging it at the same time.
4. The MOSFET turned off, like Step 2, L_1 discharges, and C_1 charges. At the output side, L_2 released the stored energy through the diode, completing the circuit through R_o .
5. Step 3 and step 4 continue to alternate as the converter operates.

3. Proposed I-V Curve Tracer Design

3.1 Converter Design

The choice of components value was selected in close reference to [4], followed by computer simulation and then physical testing to fine-tune as needed.

The component values were initially estimated using the following formula. [4]

$$L_1 = \frac{R_{mp}D}{f_s \Delta I_i} \quad (4)$$

$$L_2 = \frac{gain \cdot R_{mp}D}{f_s \Delta I_o} \quad (5)$$

$$C_1 = \frac{D(1-D)}{f_s \Delta V_{C1} R_{mp} \cdot gain} \quad (6)$$

$$C_{in} = \frac{4\Delta I_i}{\pi^3 f_s \Delta V_i} \quad (7)$$

where,

$$D = \frac{gain}{gain+1} \quad (8)$$

$$gain = \sqrt{\frac{R_o}{R_{mp}}} \quad (9)$$

$$R_{mp} = \frac{V_{mp}}{I_{mp}} \quad (10)$$

Table 2 Summarization of calculation values.

Parameters	Values
Switching frequency – f_s	100 kHz
Maximum input current ripple – ΔI_i	0.1 (10%)
Maximum input voltage ripple – ΔV_i	0.02 (2%)
Maximum output current ripple – ΔI_o	0.1 (10%)
Maximum voltage ripple at C_1 – ΔV_{C1}	0.1 (10%)
Input capacitance – C_{in}	6.46 μ F

Ćuk Capacitor – C_1	3.175 μ F
Input inductance – L_1	338.10 μ H
Output inductance – L_o	436.49 μ H
Load resistor – R_o	10 Ω

The component values were calculated based on the maximum operating values of the proposed design of V_{oc} and I_{sc} of 60V and 10A respectively for a single cell single module solar panel. Simulation was performed in MATLAB/Simulink for fine-tuning, e.g. minimizing the ripples at the input. Table 3 presents the finalised component values.

Table 3 Simulation-tuned circuit components values.

Parameters	Values
Input capacitance – C_{in}	47 μ 63V
Ćuk Capacitor – C_1	3.3 μ F 250V
Input inductance – L_1	380 μ H 10.5A
Output inductance – L_o	380 μ H 10.5A
Load resistor – R_o	20 Ω 30W

The circuit was constructed on a protoboard. In contrast to the simulation, it was found that the circuit was unable to scan beyond maximum power point (MPP) of 80W when using the 20 Ω resistor. This is likely due to voltage oscillation observed at the MOSFET gate from high di/dt during switching. Significant switching noise was also present in the ground wire, which was likely due to the absence of direct grounding plane on a protoboard. This problem was later mitigated using an output resistor of 150 Ω . The compromise of using a higher resistor value is higher heat dissipation in the circuit.

3.2 System setup

Figure 2 depicts the overview design of the Ćuk converter based I-V Curve tracer.

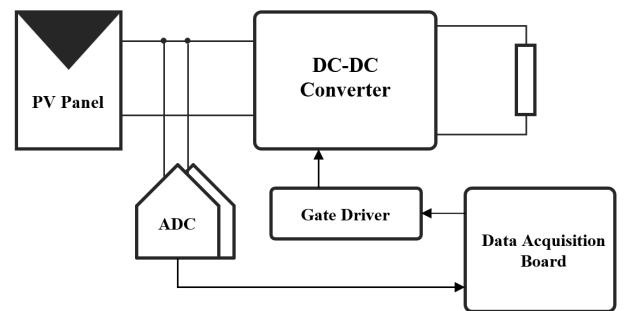


Figure 2: I-V Curve Tracer Circuit Design Block Diagram

The tracing circuit consists of a Ćuk DC-DC converter implemented with a power MOSFET, IRF540NPBF as the switching device. The MOSFET is rated at 100V and 33A, featuring fast switching speed which is well suited for the application. The switching of the device is controlled by the PWM output of a data acquisition board through a gate driver, IR2113PBF. The MOSFET and gate driver are configured in low side configuration for simplicity of the circuit.

At the input side of the circuit, voltage and current are measured for the I - V curve tracing. The measurements were taken by reading the analogue values through MCP3008 10bit analogue to digital chip (ADC). The ADC chip chosen is rated at 5V maximum input voltage; therefore, a potential divider circuit is designed with ratio of 1 to 12 for measurement of up to 60V. ACS712 hall-effect-base linear current sensor module rated at 20A was chosen for current sensing.

Raspberry Pi 4 was chosen as the data acquisition board, though any microcontroller capable of voltage and current sensing, and PWM signal generation would suffice. The data acquisition board is used for PWM generation for the switching of the converter at 100kHz and reading the measurement of voltage and current from the ADC chip through SPI communication protocol.

The measured I - V curve data is processed programmatically to smoothen the curve by removing outlier values measured due to noise or incorrect readings from the ADC chip. This is accomplished with a simple filtering algorithm. The algorithm compares the current data point with the values immediately before and after it. If the current data point exceeds predefined upper or lower boundaries, it will be discarded, as shown below:

$$\text{Upper bound} = \text{Max}(\text{val}_1, \text{val}_2) \times (1 + \text{error}) \quad (11)$$

$$\text{Lower bound} = \text{Min}(\text{val}_1, \text{val}_2) \times (1 - \text{error}) \quad (12)$$

Equation (11) and (12) define the upper and lower boundary calculations. Max and Min are the larger or smaller of the two values respectively. val_1 and val_2 are the data points before and after the current one, while error is a value between 0 and 1 defining the tolerance for outliers. The data is further refined by taking the average of few data points recorded on each step of a single duty cycle change.

The PWM duty cycle of MOSFET switching increases from 0% to 100% which reflects on the I - V curve as a movement from open-circuit voltage (V_{oc}) to short-circuit current (I_{sc}). The varying of duty cycle across the curve is not fully linear. Figure 3 (a) and (b) depict the change of duty cycle with respect to the I - V curve measured.

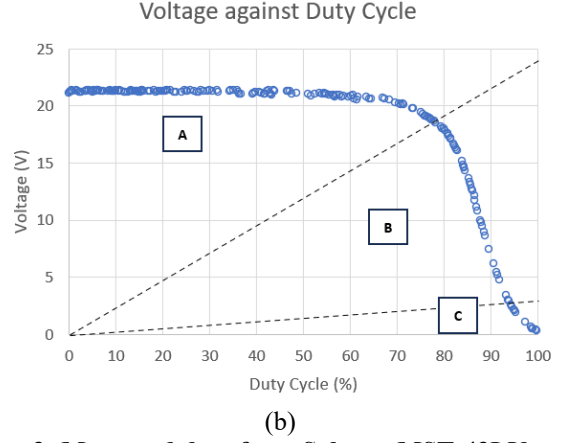
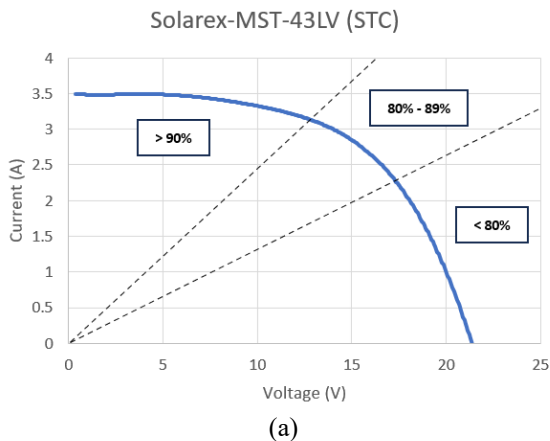


Figure 3: Measured data from Solarex-MST-43LV
(a) Example I - V curve at STC with estimated duty cycle range
(b) Graph of voltage against duty cycle

Figure 3 (a) shows the I - V curve measured from Solarex-MST-43LV PV module at STC with the estimated ranges of duty cycles, while Figure 3 (b) shows the graph of voltage against duty cycle. From Figure 3 (b), it is observed that in Region A, the voltage change is minimal from 0% duty cycle to around 75% duty cycle. Whereas in Region B, the change in voltage is more sensitive to duty cycle, i.e. at 80% to 90% duty cycle, the measured voltage changes from around 17V to 6V. This phenomenon is less severe in Region C as the duty cycle approaches 95%.

In this manner, if the duty cycle was to vary linearly, large amount of data points may not be recorded at the MPP region (~80% to ~89%). Another approach is to vary the duty cycle with linear gain; however, this leads to several issues. a) The data points at V_{oc} region will reduce. b) Asymptote is observed at duty cycle close to 100%, which implies that it is impossible to get a complete scan of the curve. c) Due to the asymptote, large number of points will be gathered in the I_{sc} region. Besides, this causes the circuit to linger at the short-circuit current region for a proportionally long duration, leading to unnecessarily high stress on the circuit components.

A feasible solution to overcome this issue is to combine both methods in each of the regions. With reference to Figure 3, Region A and C has data points that exhibit a linear change in duty cycle, while Region B has data points with linear change in gain. Programmatically, a linear change in duty cycle indicates that the duty cycle is incremented in a fixed linear step. While a linear change in gain means that the gain is increased by fixed linear step. The duty cycle at a specific gain can be calculated using Equation (8). The fixed duty cycle step and fixed gain step was set to 0.2% and 0.04 respectively on each iteration with 0.01s of sleep time in between.

The proposed solution yields an even distribution of data points across the I - V curve as seen in Figure 3. Each complete I - V curve scan takes approximately 6.8 seconds to complete. The scanning time can be reduced by increasing the duty cycle and gain steps or reducing the sleep time. Effectively, either

the number of data points collected, or the accuracy has to be sacrificed.

Figure 4 illustrates the overview of the system. The additional SPI chip select pin, and the Wi-Fi capability of Raspberry Pi 4 was utilized for an TFT display and connectivity to a web server with cloud-based database storage.

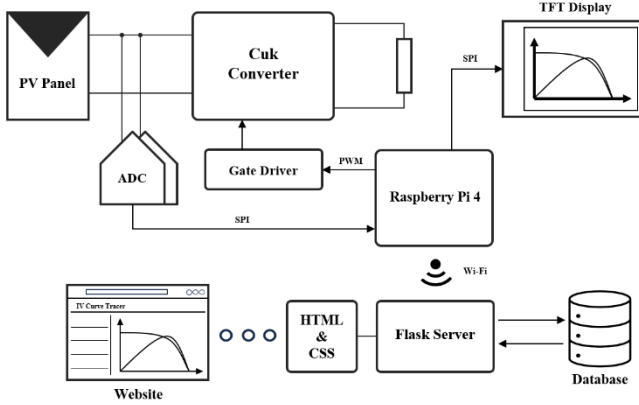


Figure 4: Proposed I - V Tracer Block Diagram

The additional TFT display allows quick analysis of the solar cell characteristics by visualizing the I - V and P - V curve. Key parameters such as open circuit voltage, short circuit current, maximum power point, and fill-factor are also shown. Furthermore, the data measured by the I - V curve tracer can be transmitted over to a Flask-based web server hosted on Render with Google Firebase integration for database storage. The website serves as an interface for accessing data stored in the database, displaying the I - V curves, and allowing downloads of the data in the form of .csv files for further analysis.

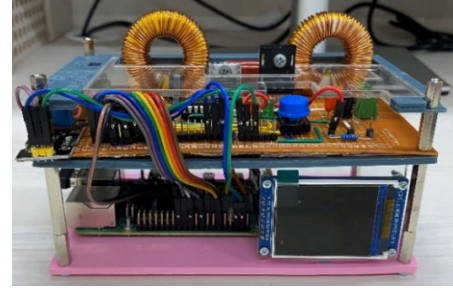
The server is built using the Flask framework in Python, serving as the centre hub for data flow between the tracer device, database, and frontend. The website's frontend is developed using NiceAdmin template from BootstrapMade.

Data transfer between the prototype and the web happens through sending formatted JSON strings directly to the server address via Wi-Fi. All subsequent data processing is managed by the Flask server, independently from the tracer device. Upon receiving the JSON string, the server unpacks the data and put it in a .csv file for storage in the cloud database. Figure 4 depicts how the data flows between each block. This prototype does not incorporate encryption for data security, it was designed solely as a proof of concept.

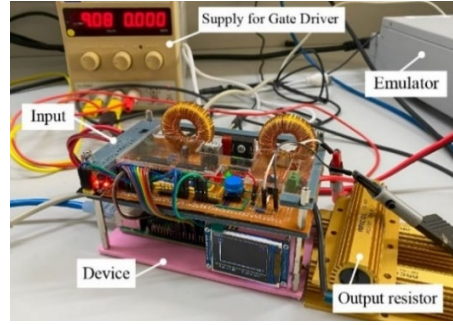
4. Device Implementation

4.1 I - V Curve Tracer

Figure 5 shows the final prototype of the I - V curve tracer. The device is structured in two levels: the top level features the I - V curve tracer circuit constructed on a protoboard, while the bottom level houses the Raspberry Pi 4 and the TFT display.



(a)



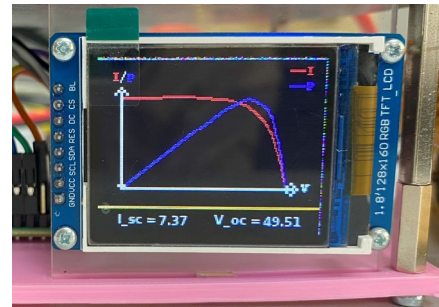
(b)

Figure 5: Hardware of I - V Curve Tracer Prototype

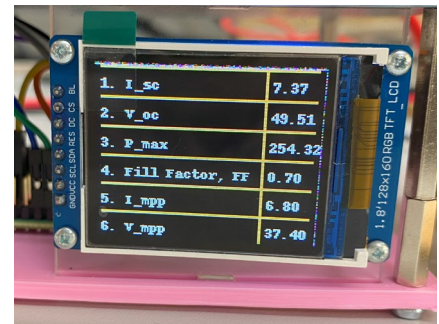
(a) The Device alone

(b) Testing Setup with Emulator

The I - V curve tracer circuit at the top level consist of the Cuk converter circuit, sensor circuits discussed in the previous section, and several male header pins serving as interfaces for power supplies and to facilitate input/output data connections to the Raspberry Pi. A button is included for controlling the device, along with two LED indicators. A single click of the button switches between two pages on the TFT display, while a long press initiates the I - V curve scanning process.



(a)



(b)

Figure 6: TFT Display

- (a) Page 1 of user interface
(b) Page 2 of user interface



Figure 7: Screenshot of the I - V Data Webserver

The TFT display shows key information about the I - V curves across two pages. Figure 6 (a) shows the first page, where the I - V curve and P - V curve are displayed on the same axis. This provides a quick visual overview of the curves, allowing for an assessment of the solar panel's characteristics. At the bottom, the open-circuit voltage and short-circuit current are displayed for quick reference. The second page, shown in Figure 6 (b), provides more detailed information, including key parameters such as open-circuit voltage, short-circuit current, fill factor, and the voltage, current, and power at the maximum power point.

The website is hosted live on Render as a proof of concept. It is fully functional, capable of receiving data from the tracer device and retrieving data from the database. With reference to Figure 7, on the left of the website is the collections of measured I - V curves stored in the database. By clicking on anyone of the collections, the respective I - V curves will display on the right. The data can be downloaded in the form of .csv file to local computer for further data analysis.

The I - V curve measurement data is stored as '.csv' files within Google Firebase storage, providing remote, real-time accessibility to the scanned I - V curves. While Firebase is user-friendly and suitable for simple prototypes like this one, it may not be ideal for large-scale commercial applications due to its high pricing. Alternatively, other cloud storage solutions such as AWS S3 or Google Cloud Storage could offer better scalability and cost efficiency for larger deployments.

4.2 Data Analysis

The I - V curve tracer prototype was tested by tracing selected PV models using Chroma 62100H-600S programmable solar array simulator. The prototype was tested at its rated maximum power limit of 250W to demonstrate its capability.

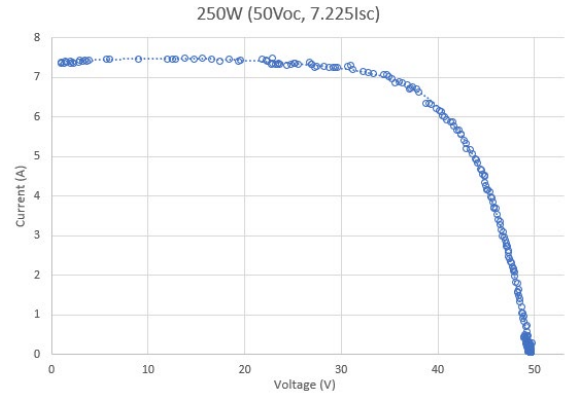
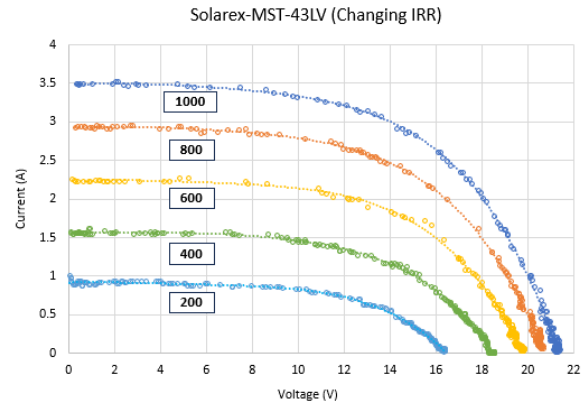


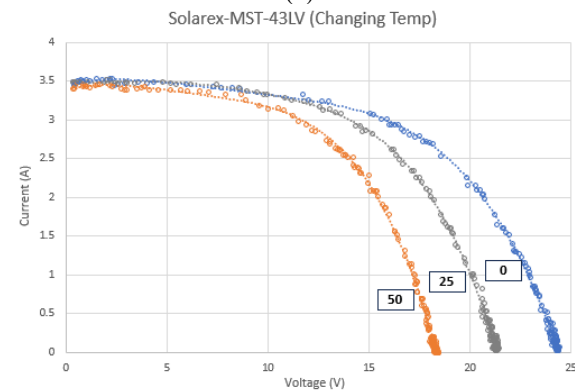
Figure 8: Measured I - V curves of Sandia Model at 250W MPP

The SAS modelling featured in the emulator was used with an V_{oc} of 50V and I_{sc} of 7.225A. The Sandia model was configured for standard test conditions (STC) with a fill factor (FF) set to 0.68. Figure 8 shows the data points collected. The scanned result is acceptable with measured $V_{oc} = 49.51V$, $I_{sc} = 7.37$, $MPP = 254.32$, and $FF = 0.7$.

Three commercially available PV models were selected for testing, each representing a different power range. The models, listed from lowest to highest power, are the Solarex MST-43LV, Mitsubishi PV-MF170EB4, and Sanyo H168. Each model was simulated and has its output traced under varying temperature and irradiance conditions. This was accomplished using the Chroma 62100H-600S Solar Array Simulator.



(a)



(b)

Figure 9: I - V Curves of Solarex-MST-43LV measured using the proposed I - V curve tracer.

(a) Irradiance Variation at 25°C

(b) Temperature Variation at 1000W/m²

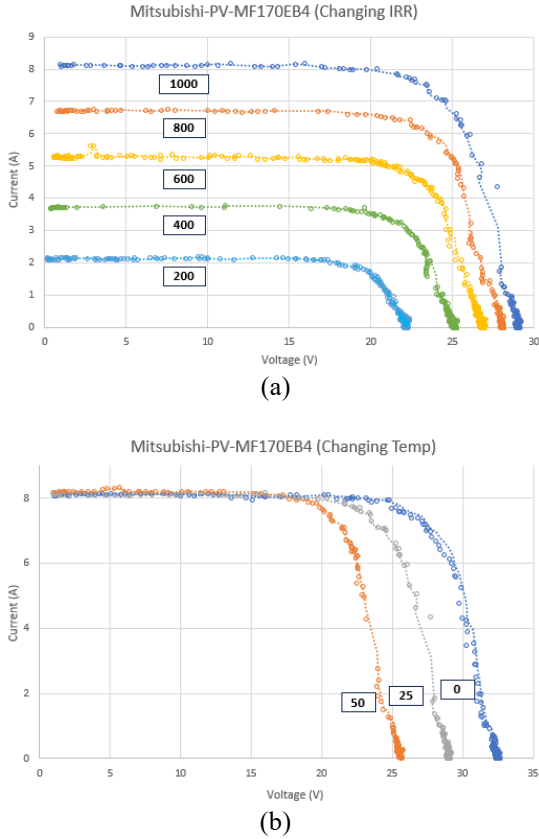


Figure 10: I - V curves of Mitsubishi PV-MF170EB4 measured using the proposed I - V curve tracer

(a) Irradiance Variation at 25°C

(b) Temperature Variation at 1000W/m²

Figure 9 (a), (b) and Figure 10 (a), (b) plotted the data points captured by the prototype on varying irradiances and temperatures respectively. The trend of I - V curves correlates with a typical I - V curve under varying irradiance and temperature conditions. With increasing irradiance, the short-circuit current increases in linear steps and some minor change in the open-circuit voltage. Likewise, with increasing temperature, the open-circuit voltage reduces in linear steps, while the short-circuit current stays the same.

Through visual observation of the data points plotted, at lower power (around 43W) with Solarex PV module, more consistent and smoother curve can be observed, at higher power (around 170W) with Mitsubishi PV module, lesser concentration of data points exists between the MPP and open-circuit voltage. This hypothesis is that, at higher power, there exist significant fluctuation in the measured data which are then eliminated by the filtering algorithm. The existence of fluctuations in the measured data could be attributed to the high switching noise observed throughout the circuit.

Table 3 Summarisation of Root Mean Square Error

Irr (W/m²) T (°C) RMSE (A) RRMSE (%)

Solarex-MST-43LV			
200	25	0.0651	8.9494
400	25	0.0560	4.3570
600	25	0.0951	5.2657
800	25	0.2029	8.8020
1000	25	0.1469	5.2667
1000	50	0.1338	4.8000
1000	0	0.1636	5.8670
Mitsubishi PV-MF170EB4			
200	25	0.2222	11.5378
400	25	0.3229	9.4818
600	25	0.3360	7.0222
800	25	0.3352	5.4917
1000	25	0.3531	4.7824
1000	50	0.3454	4.6778
1000	0	0.3454	4.6780

Table 3 detailed the Root Mean Square Error (RMSE) and Relative Root Mean Square Error (RRMSE) values calculated for each of the test cases with the equations (13) and (14).

$$RMSE = \sqrt{\frac{1}{n} \sum (I_{measured} - I_{reference})^2} \quad (13)$$

$$RRMSE = \frac{RMSE}{\text{Mean of Reference Current Values}} \times 100\% \quad (14)$$

where, n is the total number of data points measured.

The RMSE values represent the absolute error in Amperes of the measured current to the reference I - V curve. RRMSE values, however, shows the significance of the error to the scale of the current measurement.

The average RMSE and RRMSE at the lower power PV model is 0.1233A and 6.1868% respectively, while at higher power PV model, the average is 0.3229A and 6.8102% respectively. From the trend, it was observed that as the power goes up, the RMSE also values increase. However, considering RRMSE values, the errors are least significant as the power goes up. Another observation is that at lower power, both RMSE and RRMSE values are fluctuating without a clear trend, but the device performs are more stable at higher power PV models.

The comparison between the measured data points and the reference curve is depicted in Figure 11 (a), (b), and (c). From the graphs, it was observed that even though data in Figure 11 (a) has RMSE of 0.0651, but this error is more significant to the scale of the plot when compared to data measured in Figure 11 (b) with RMSE of 0.3531. Moreover, observing graph in Figure 11 (c) with the highest RRMSE value and relatively close RMSE value to the other plots, the data points collected are further away in scale relative to the reference curve, indicating poor measured data comparing to the reference curve.

Overall, the RMSE and RRMSE values are not ideal for the application of I - V curve tracing, but the results are still acceptable given the cost of this prototype at RM140.00 (Roughly £25). Most of the RRMSE values are below 10%, indicating no major discrepancy between the measured I - V curve and the ideal one.

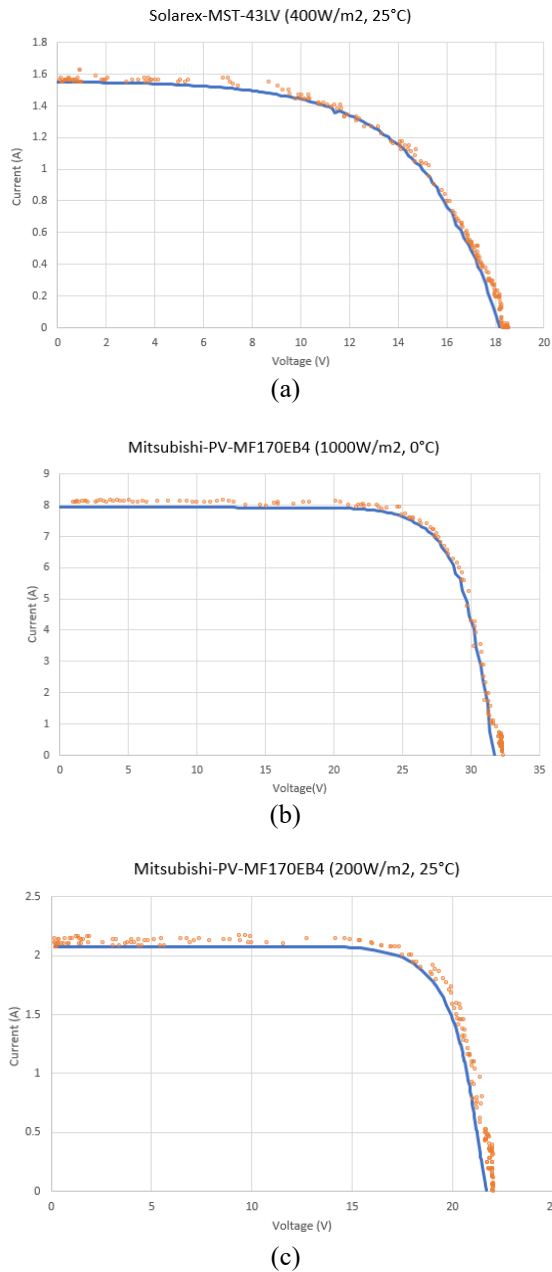


Figure 11: Comparison between the measured and reference I - V datapoints

- (a) Solarex-MST-43LV at 400W/m^2 , 25°C
- (b) Mitsubishi-PV-MF170EB4 at 1000W/m^2 , 0°C
- (c) Mitsubishi-PV-MF170EB4 at 200W/m^2 , 25°C

In addition, Sanyo-H168 PV model was tested to show the limitation of the I - V curve tracer prototype. The prototype was built with the capability to measure up to $60V_{oc}$ only. This is because, the potential divider circuit was made to supply a

maximum of 5V to the ADC chip at 60V with accordance to the maximum voltage rating of the chip. When testing with curves beyond 60V, the ADC chip reading saturates as seen in Figure 12 (a) and (b).

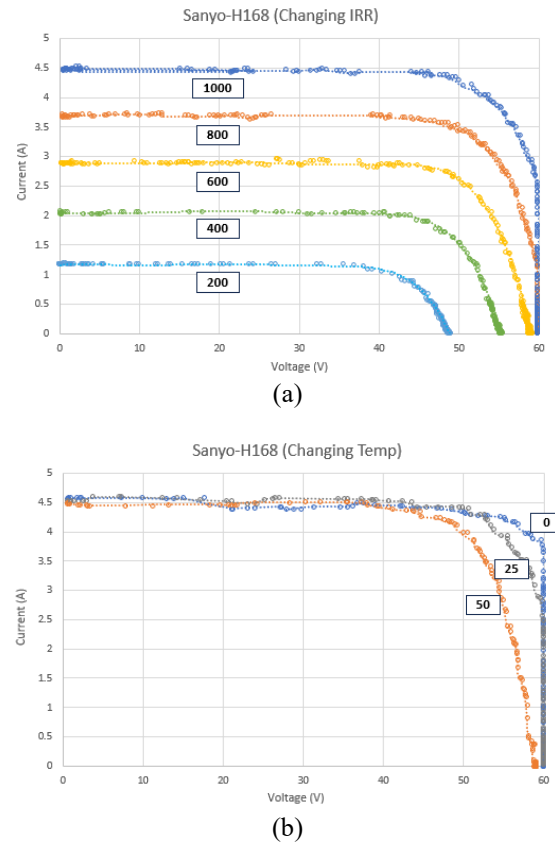


Figure 12: I - V Curve of Sanyo-H168 measured using the proposed I - V curve tracer.

- (a) Irradiance Variation at 25°C
- (b) Temperature Variation at 1000W/m^2

Note that it is not recommended to scan curves beyond 60V, as testing too far beyond the rated V_{oc} may eventually destroy the ADC chip. Likewise, scanning curves beyond the rated 250W MPP will induce excessive stress on the circuit. At high power level, the MOSFET heats up rapidly, reducing the ability to conduct higher currents, resulting in failure of the component. Other components such as the diode can also be affected due to high reverse current surge in the event of MOSFET failure.

5. Conclusion

The concept of an internet-enabled I - V curve tracer device was successfully implemented with a focus on cost-effectiveness. The prototype demonstrated that incorporating internet connectivity for remote monitoring and data access is feasible and practical for real-world applications. By establishing a simple webserver, the system allows for remote, real-time monitoring of a solar cell's I - V characteristics and provides access to historical data. This integration would significantly simplify the solar cell's health check and performance analysis procedure.

Due to significant noise present throughout the circuit, the collected data was processed programmatically to smoothen the curve by removing outlier values. Further refinement was achieved by averaging multiple data points recorded at each step of a single duty cycle change.

The proposed solution for evenly spread data points collection was by combining the use of linear duty cycle change and linear gain change in different region of the I - V curve. Each complete scan takes approximately 6.8 seconds. The scanning time can be reduced by either sacrificing the number of data points collected, or the accuracy.

With emphasis on cost-effectiveness, the prototype rounds up to a total cost of RM140.00 or approximately £25, excluding the data acquisition board. The design is adaptable, allowing for the use of any data acquisition board or microcontroller capable of providing a stable 100kHz PWM signal. Despite the prototype being low budget, it produced satisfactory results. For a low-power PV module, the average RMSE was 0.1233A, with a RRMSE of 6.1868%. For a high-power PV module, the average RMSE increased to 0.3229A, with a RRMSE of 6.8102%.

Overall, the proposed device achieved an average RMSE and RRMSE of 0.2848A and 6.4985% respectively. While the RRMSE value is below 10%, which is within acceptable limits, it is not ideal for applications requiring precise I - V curve tracing.

It is worth noting that the current prototype was assembled on a protoboard which introduced significant noise and signal integrity issues that negatively impacted the overall performance. Without a short and direct grounding on a protoboard, switching noise from the MOSFET propagates throughout the circuit, interfering the accuracy of measured signals. From the results discussed in the previous section, the traced I - V curves were not consistently stable. There exist missing data points on the curves resulting from significant fluctuations of measured values that are being filtered by the program.

The device can be further improved by reducing the inconsistency in measured I - V values. Implementing a printed circuit board (PCB) design with a proper ground plane would significantly reduce noise and improve overall performance. Additionally, the use of a more organized routing strategy and optimized component placement, along with surface-mount devices (SMDs), would result in a compact layout that further enhances signal integrity and minimizes interference.

To conclude, the prototype successfully meets the project's objectives, demonstrating how IoT technology can be integrated into an I - V curve tracer device. Future improvements could be made to achieve a more precise I - V curve tracer while maintaining the low cost.

6. Acknowledgement

The authors would like to thank University of Southampton Malaysia and the Ministry of Higher Education of Malaysia for providing the facilities and financial support through the Fundamental Research Grant Scheme (FRGS), grant number FRGS/1/2022/TK08/USMC/03/2.

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